

6.4 Uncertainty in sample statistics [↕]

□ Watch: Visualising variability - (part 4) [25.35 mins]

Watch the following video, which explains what is meant by statistical inference. The confidence interval is also defined and prediction intervals are introduced.

This video is useful as it assists your understanding of how to convey the uncertainty that arises in statistical inference and predictive analytics.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138647?wrap=1>

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Download the following slides for note taking: [ICT702 Visualising variability_\(part 4\)](https://learn.usc.edu.au/courses/35242/files/3138667?wrap=1)
<https://learn.usc.edu.au/courses/35242/files/3138667?wrap=1>

Statistical inference is the process of drawing conclusions or making estimates about a population (or characteristic thereof) based on a sample or subset.

There would understandably be a level of error or deviation between the results from a sample of the population and the results that would be obtained if the whole population was able to be included in a study. As a result, there will be uncertainty in how much the sample proportion represents the whole population. It is important to convey this inherent uncertainty to the audience, for without doing so might result in the audience developing an inappropriate level of

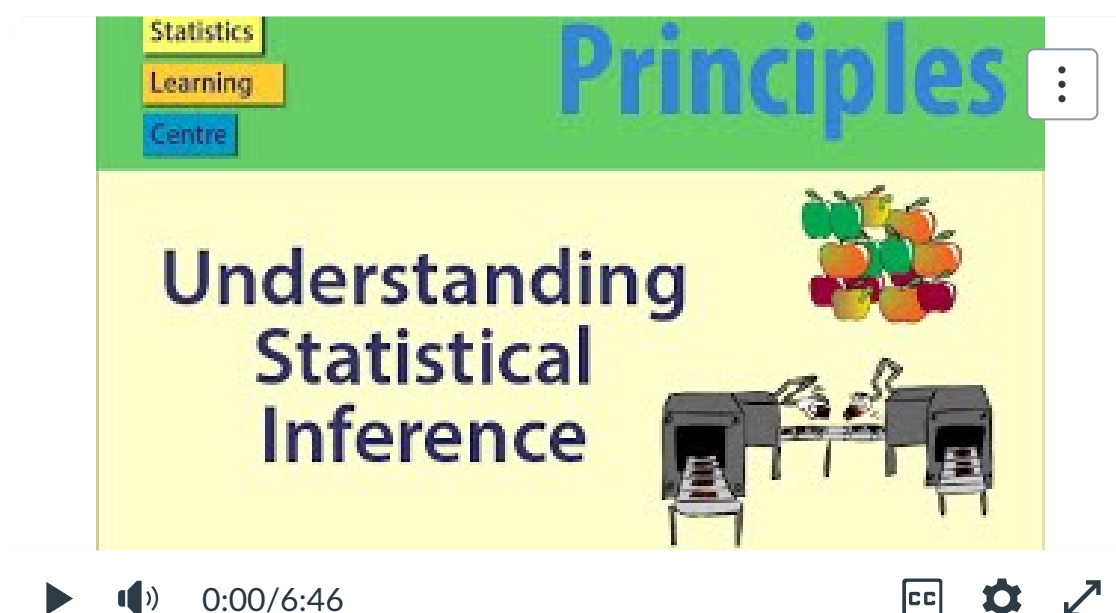
confidence in the estimates. Therefore, visualisations need to help the audience understand the estimates as intervals.

□ Watch: Understanding statistical inference [6.45 mins]

Watch the following video, which explains what statistical inference is. This video is useful, because statistical inference is one of the harder statistical concepts to comprehend.

As you watch, consider:

- How can the understanding gained from this video be used to improve the data stories that you share?
- Why is understanding statistical inference important?



Source: © Dr Nick's maths and stats (2015).

Displaying a confidence interval on a mean

It should not be expected that the sample mean will provide the exact value of the population mean. As a result, a “confidence interval” should be computed. This can be done by subtracting a value known as the “margin of error”.

Confidence interval

This represents how good an estimate is. In this case, its purpose is to provide information regarding how close the sample mean might be to the population mean.

Margin of error

The margin of error allows the level of unpredictability in data and research outcomes to be gauged. It is dependent on three factors:

- sample size
- how variable the sample values are
- how much confidence can be claimed that the population mean lies within the interval.

Intuitively, larger sample sizes result in smaller margins of error; with more data, it should be possible to better estimate the mean. Equally, when sample standard deviation increases, so does the margin of error (as data values have more variation, it is more difficult to estimate the mean); and when the required confidence level increases, so does the margin of error. When it is important to be more confident in an interval, then the width of the interval must be increased. Usually, levels of confidence are given as 95% or 99%.

Displaying confidence interval on a proportion

Similarly, the degree of uncertainty in a sample proportion can also be expressed by a confidence interval that has been formed from adding or subtracting a margin or error. In this case, the purpose of the confidence interval is to provide information regarding how close the sample proportion might be to the population proportion. A common level of confidence is 95%.

In this case, the margin of error is dependent on:

- sample size
- sample proportion
- how much confidence can be claimed that the population proportion lies within the interval.

□ Watch: Understanding confidence intervals [4.02 mins]

The following video provides a useful explanation about the concept of confidence intervals.

As you watch consider:

- How can this added understanding about confidence intervals be used to improve your data visualisations?
- Why is this understanding important?



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